AI- Based Flash Flood Prediction based on Rainfall and River Level Data: A Bibliometric Study and Analysis using VOSviewer

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Abstract

Flash floods are emerging as a major threat to hilly areas in India especially Uttarakhand and bringing about severe social and economic issues. The uncertainty and the increased occurrence of such incidents has been necessitating better prediction mechanisms. Deep learning and artificial intelligence in general could improve the precision and predictive time of forecast. The software VOSviewer is used to map publications found in the Google scholar between 2019 and 2025 in order to map publications published in the past five years.

With the effective keywords flash flood, deep learning, and AI prediction, the study outlines the research topics of concern and demonstrates aggressive research communities, leading authors, and interests. The given research is among the first to establish research trends and collaborations in the context of AI-based flash flood forecasting because before that time there were extremely limited bibliometric works on the topic in hilly areas. The results provided would give a global perspective of the great research teams, which would influence further developments in disaster management and early warning systems.

Keywords: deep learning, flash flood, bibliometric analysis, VOSviewer, mountainous region.

Introduction

Flash floods are sudden and violent rises in the water levels and are normally caused by heavy rainfalls, bursting of glaciers or cloud bursts. These floods are more disastrous in the hilly regions of Uttarakhand where the ruggedness of the topography and the weakened landscape enhances the run off and causes the local and instantaneous floods. They cannot be predicted and rather sudden in their occurrence, and with this lack of predictability, these events leave devastation in the management of disasters since they are likely to leave massive destruction of structures and lives lost.

Artificial intelligence and, to be more precise, the techniques of deep learning have been the efficient tools that are used worldwide to assist in the prediction of floods. The models can be applied to enhance early warning and reduce risks because of identifying subtle patterns in volumes of climate and hydrological data.

Although many studies have been conducted on the use of AI in floods forecasting, few studies have conducted research reviews of how AI can be used in mountainous regions.

Bibliometric analysis is an orderly procedure of measuring the development of research, trend and networks. It will target providing the first large-scale bibliometric map of the existing research of AI-based flash flood forecasting in mountainous areas with the assistance of VOSviewer software. It is conducted to identify the primary contributors, theme groups, and areas of omission hotspots to future studies.

Materials and Methods

The current research conducted a bibliometric analysis on articles found in Google Scholar and other prestigious journal-based sources of research articles published in the past three years, that is, 2019- 2025. Search terms were made up of a combination of the search terms as flash flood, deep learning, flood prediction and other similar terms. Significant data in each article were extracted and displayed in an organized CSV file, comprising of the paper title, the names of the authors, the date of publication, the journal/source, the abstract, the keywords, the document identifiers, the citation entries, and total citations. All data was checked as correct and complete manually. The final set was a group of ten significant articles which were dedicated to the innovative approaches to flood forecasting. Bibliometric network and mapping visualizations were performed with VOSviewer (version 1.6.20) in order to explore the co-occurrence of key-words, identify prolific journals, analyse cited references and track thematic patterns in the emerging literature regarding deep learning in flood risk forecasting.

Results and Discussion

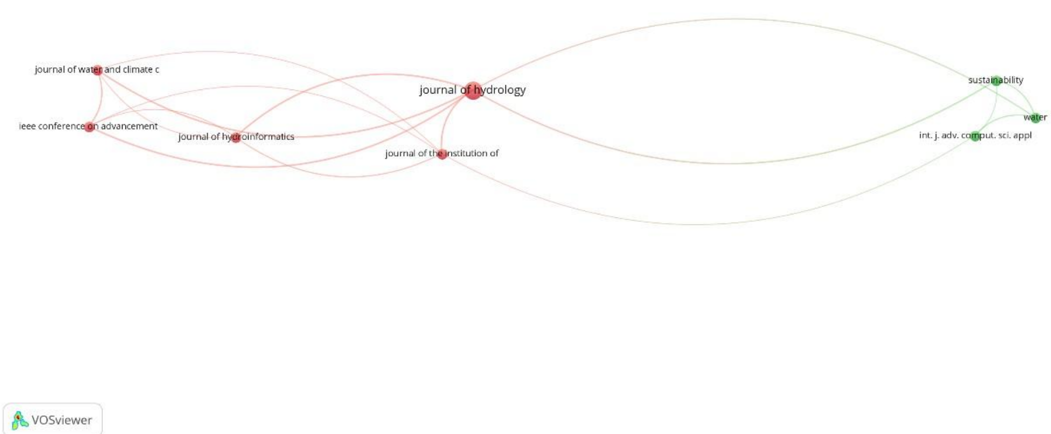
- **Bibliometric Analysis of Sources**

The review of the ten selected articles revealed that articles published in the journals like Water, Journal of Hydrology, and International Journal of Advanced Computer Science and Applications included research on deep learning flood forecasting on regular basis. It was these sources which published the simple methodologies and the new AI-hydrological development of models. These core journals and their role in the field of research and collaboration between disciplines was pointed out with the help of citation data.

Table 1. The eight leading journals featuring research on deep learning-based flood prediction.

Selected	Source	Documents	Citations	Total link strength
☑	journal of hydrology	3	324	21
☑	ieee conference on advancements in c...	1	10	9
☑	journal of the institution of engineers	1	18	9
☑	journal of water and climate change	1	193	9
☑	journal of hydroinformatics	1	20	8
☑	sustainability	1	177	6
☑	water	1	104	6
☑	int. j. adv. comput. sci. appl.	1	29	4

Figure 1. Bibliographic coupling Network of sources.



As Figure 1 shows, journals that present flood prediction studies based on deep learning are bibliographically connected to each other. These circles are the number of journals and the width of the circle is the total number of citations. Colors denote groups of journals that are likely to cite papers of each other or there is similarity in terms of thematic or methodological content. Publications like Water and Journal of Hydrology are central hubs and there is very strong interconnection between various clusters.

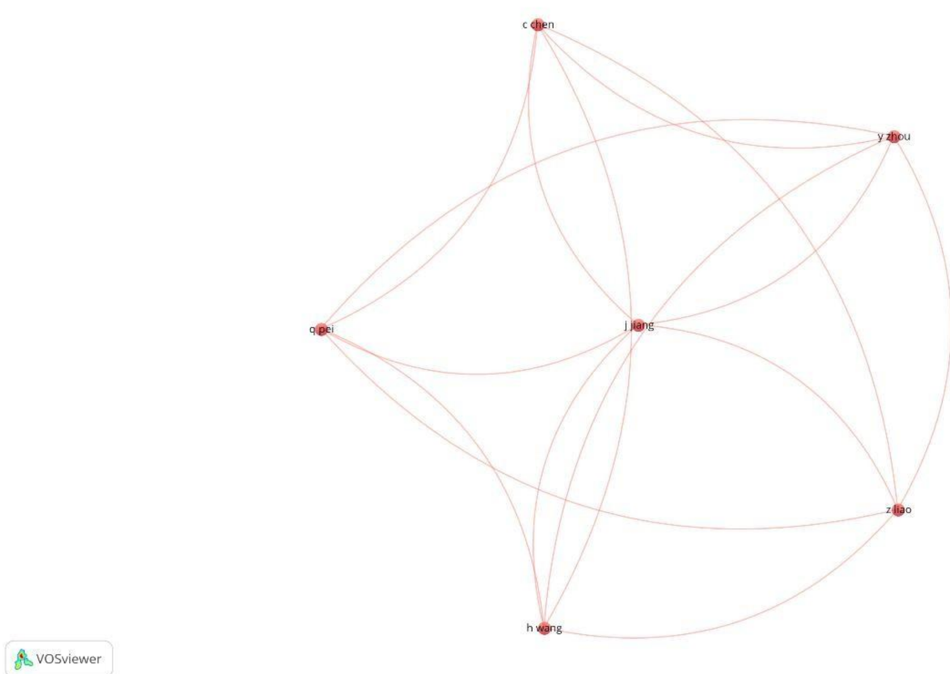
- **Bibliometric Analysis of Authors**

  The corpus reflected groupings of authors that contributed to the confluence of hydrology and machine learning multiple times. It should be mentioned that the fusion of other authors was common, which could be attributed to the need to combine knowledge to develop and test the flood prediction models. The co-authorship patterns were also used to show that collaborations between applied hydrology and data science experts were formed.

Table 2. The Most Important Authors in Studies of Deep Learning for Flood Prediction

Selected	Author	Documents	Citations	Total link strength
☑	c chen	1	102	5
☑	h wang	1	102	5
☑	j jiang	1	102	5
☑	q pei	1	102	5
☑	y zhou	1	102	5
☑	z liao	1	102	5
☑	cc pain	1	198	3
☑	dj mehta	1	177	3
☑	f fang	1	198	3
☑	hm azamathulla	1	177	3
☑	im navon	1	198	3
☑	j chen	1	20	3
☑	kv sharma	1	177	3
☑	mhm ali	1	29	3
☑	r hu	1	198	3
☑	sa asmai	1	29	3
☑	t yu	1	20	3
☑	t zhang	1	20	3
☑	v kumar	1	177	3

Figure 2 displays the co-authorship network for researchers who focus on flood prediction with deep learning techniques.



The co-authorship graph of top writers is plotted in Figure 2. Greater productivity is represented by bigger circles and collaborative relationships by the connecting lines between nodes. The figure indicates that international collaboration is common in research in this area, particularly among Asian and European institutions.

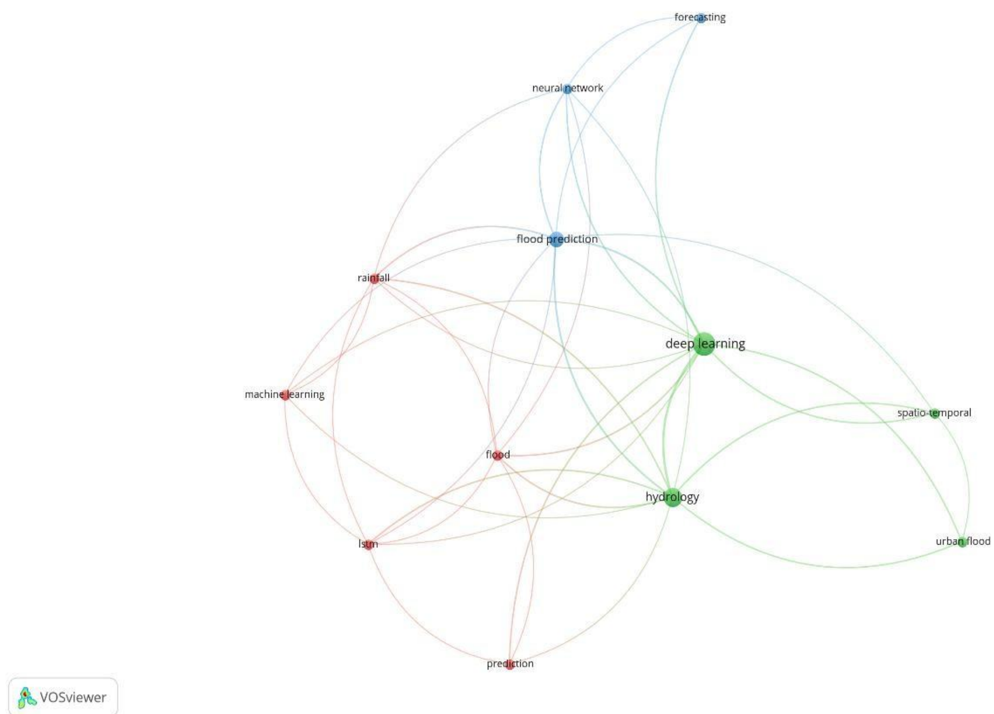
- **Keyword Co-occurrence Analysis**

  The VOSviewer analysis showed that the keywords "flash flood," "deep learning," "LSTM," and "hydrology" were the most strongly linked and made up the main nodes of the research network. Flood-risk assessment schemes are becoming broader in scope and are constantly being improved by increasing reliance on data, as can be plainly seen from the growing popularity of clusters such as uncertainty estimation, satellite image processing, and spatial analysis.

Table 3 shows the major phrasal word theme identified, which represents research on flood prediction using deep learning (DL).

Selected	Keyword	Occurrences	Total link strength
☑	deep learning	9	23
☑	hydrology	6	21
☑	flood prediction	4	15
☑	flood	2	9
☑	rainfall	2	9
☑	lstm	2	8
☑	neural network	2	8
☑	spatio-temporal	2	6
☑	machine learning	2	5
☑	prediction	2	5
☑	urban flood	2	5
☑	forecasting	2	4

Figure 3. Visualization of the Keyword Co-occurrence Network.



VOSviewer generated a co-occurring keywords network that is depicted in Figure 3. The color-filled clusters stand for different themes with the node size replicating keyword occurrences. The main trend of AI and hydrological modeling merger is indicated by the clusters around "remote sensing," "rainfall forecasting," and "neural networks". The color-filled clusters stand for different themes with the node size replicating keyword occurrences.

- **Bibliometric Examination of Co-Citations**
  Co-citation mapping located among others Water and Journal of Hydrology as the most commonly co-cited thus underlining their leading position as the main authors of technical guidance and basic research. The most cited papers quite often referred to the same old studies more than once which suggests that the discipline was adopting a similar methodological approach. This very resemblance signifies the establishment of the best practice and the mutual concurrence of scientific standards for AI-assisted flood forecasting.

Table 4. Most Frequently Co-Cited Sources

Selected	Source	Citations	Total link strength
☑	hu r 2019	6	11
☑	chen c 2022	4	8
☑	kumar v 2023	4	7
☑	ali mhm 2022	3	6
☑	gude v 2020	3	6
☑	nayak m 2022	3	6
☑	pathirana d 2019	2	4
☑	sankaranarayanan s 2020	2	4
☑	shao y 2024	2	4

Figure 4. Co-Citation Network for Cited Sources.

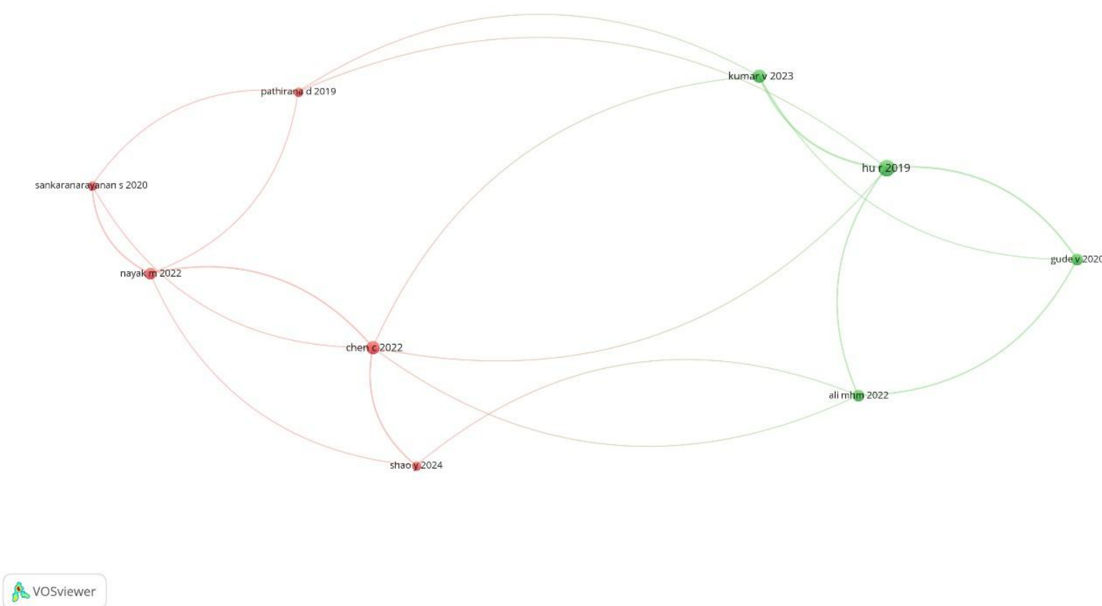


Figure 4 illustrates the co-citation network of the most reputable journals in the area of flood prediction using deep learning methods. The co-cited references along with the foremost nodes such as Journal of Hydrology and Water have been highlighted by their respective locations at the center. Diverse research areas are represented through a color scheme, thereby reflecting hydrology and AI-assistive modeling as two among several others. The diameter of each node is indicative of the amount of citations received and is one of the factors that highlight the significance of these journals in this particular research domain.

Conclusion

The bibliometric analysis conducted a meticulous examination of 10 peer-reviewed articles, which were published between 2019 and 2025, that had different deep learning techniques for flood prediction as their main topic. The study named "Water" and "Journal of Hydrology" as the most referenced and "flash flood," "deep learning," and "LSTM" as the most frequently used terms in the research. It is very surprising that many studies have indicated a considerable research gap, stating that only a few studies have resorted to AI along with rainfall and river level observations for precise flash flood predictions in mountainous areas. This paper is the first one to use bibliometric methods to depict the publication of AI-based flash flood prediction research and how that has been facilitated by the VOSviewer program, thereby revealing the collaboration networks, leading sources, and new trends in the field. Later researchers might take a step further by incorporating the use of IoT sensor networks and satellite imaging in their assessment or by contrasting the coverage of Scopus and Web of Science for a more holistic view of the literature. The findings have paved the way for even more creative and location-specific methods in flash flood simulation and warning systems. The bibliometric analysis has provided insights that are very helpful in the process of developing regional policies and coordinating real-time flood monitoring in areas where data is scarce, especially in hilly regions.

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